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United States Patent Application Publication

20250285809

Kind Code

A1

Publication Date

September 11, 2025

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STRUCTURE AND METHODS OF FORMING THE STRUCTURE

Abstract

Capacitors, apparatus including a capacitor, and methods for forming a capacitor are provided. One such capacitor may include a first conductor a second conductor above the first conductor, and a dielectric between the first conductor and the second conductor. The dielectric does not cover a portion of the first conductor; and the second conductor does not cover the portion of the first conductor not covered by the dielectric.

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Family ID: 1000008619470

Appl. No.: 19/214793

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18627931 20240405 parent-grant-document US 12315683 child US 19214793

parent US continuation 17067587 20201009 parent-grant-document US 11955285 child US 18627931

parent US continuation 15948740 20180409 parent-grant-document US 10170248 child US 16237341

parent US continuation 15676597 20170814 parent-grant-document US 9941053 child US 15948740

parent US continuation 14790414 20150702 parent-grant-document US 9734949 child US 15676597

parent US division 16237341 20181231 parent-grant-document US 10804036 child US 17067587

parent US division 13214902 20110822 parent-grant-document US 9082555 child US 14790414

Publication Classification

Int. Cl.: **H01G4/30** (20060101); **H01G4/005** (20060101); **H01G4/06** (20060101); **H01L23/52** (20060101); **H01L23/522** (20060101); **H01L23/535** (20060101)

U.S. Cl.:

CPC **H01G4/30** (20130101); **H01G4/005** (20130101); **H01G4/06** (20130101); **H01L23/52** (20130101); **H01L23/522** (20130101); **H01L23/535** (20130101); Y10T29/43 (20150115)

Background/Summary

PRIORITY APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 18/627,931, filed Apr. 5, 2024, which is a continuation of U.S. application Ser. No. 17/067,587, filed Oct. 9, 2020, now issued as U.S. Pat. No. 11,955,285, which is a divisional of U.S. application Ser. No. 16/237,341, filed Dec. 31, 2018, now issued as U.S. Pat. No. 10,804,036, which is a continuation of U.S. application Ser. No. 15/948,740, filed Apr. 9, 2018, now issued as U.S. Pat. No. 10,170,248, which is a continuation of U.S. application Ser. No. 15/676,597, filed Aug. 14, 2017, now issued as U.S. Pat. No. 9,941,053, which is a continuation of U.S. application Ser. No. 14/790,414, filed Jul. 2, 2015, now issued as U.S. Pat. No. 9,734,949, which is a divisional of U.S. application Ser. No. 13/214,902, filed Aug. 22, 2011, now issued as U.S. Pat. No. 9,082,555, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to capacitors and, in a particular embodiment, to a parallel plate capacitor for storing and providing electrical energy to a variety of devices, including semiconductor devices.

BACKGROUND

[0003] Capacitors are a basic electrical element used in storing and providing electrical energy to other electrical elements. They are used in most of today's electrical and/or electronic devices and continue to expand its range of applications into new types of hi-tech devices, such as semiconductor devices, as technologies rapidly evolve. While there are a vast array of capacitors (e.g., metal oxide field effect transistor (MOSFET) capacitors) available to be used in such semiconductor devices, as the density of the semiconductor devices have exponentially and steadily increased over the years, there have been incessant and increasing demands for capacitors that are smaller in size but greater in storage capacity.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0004] FIG. 1A shows a perspective view of one illustrative embodiment of a capacitor.

[0005] FIG. 1B shows a cross-sectional view of the capacitor shown in FIG. 1A taken along line A-A'.

[0006] FIG. 1C shows a planar view of the capacitor shown in FIG. 1A.

[0007] FIG. 2A shows a perspective view of another illustrative embodiment of a capacitor.

[0008] FIG. 2B shows a cross-sectional view of the capacitor shown in FIG. 2A taken along line B-B'.

[0009] FIG. 2C shows a planar view of the capacitor shown in FIG. 2A.

[0010] FIG. 2D shows a schematic circuit diagram of the capacitor shown in FIG. 2A.

[0011] FIG. 3A shows a perspective view of yet another illustrative embodiment of a capacitor.

[0012] FIG. 3B shows a cross-sectional view of the capacitor shown in FIG. 3A taken along line C-C'.

[0013] FIG. 3C shows a cross-sectional view of the capacitor shown in FIG. 3A taken along line D-D'.

[0014] FIG. 3D shows a planar view of the capacitor shown in FIG. 3A.

[0015] FIGS. 4A and 4B show a planar view of illustrative embodiments of capacitors including four contact electrodes.

[0016] FIG. 5 shows a cross-sectional view of an illustrative embodiment of a flash memory device including a capacitor.

[0017] FIG. 6 shows an illustrative embodiment of a charge pump including multiple capacitors.

[0018] FIG. 7 shows an example flow diagram of an illustrative embodiment of a method for fabricating a capacitor.

[0019] FIGS. 8A-8G are a series of diagrams illustrating the example method shown in FIG. 7 and the structures fabricated by the example method.

[0020] FIG. 9 shows a schematic diagram of an illustrative embodiment of a system including a non-volatile memory device.

DETAILED DESCRIPTION

[0021] Techniques relating to a capacitor are provided. In one embodiment, the capacitor may include a first conductor, a second conductor above the first conductor, and a dielectric between the first conductor and the second conductor. The dielectric does not cover a portion of the first conductor; and the second conductor does not cover the portion of the first conductor not covered by the dielectric.

[0022] In another embodiment, a capacitor may include N planar conductors disposed one above the other, each of the planar conductors including at least one first portion not covered by the planar conductor disposed above thereof, wherein N is a natural number equal to or greater than two.

[0023] The foregoing embodiments are illustrative only and are not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

[0024] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

[0025] FIG. 1A shows a perspective view of one illustrative embodiment of a capacitor. FIG. 1B shows a cross-sectional view of the capacitor shown in FIG. 1A taken along line A-A'. FIG. 1C shows a planar view of the capacitor shown in FIG. 1A. Referring to FIGS. 1A-1C, a capacitor **100** may include a substrate **110**, and a stacked body **120** provided on substrate **110**. While not expressly illustrated in FIGS. 1A-1C for the sake of simplicity, a dielectric material, such as silicon oxide, may be interposed between substrate **110** and stacked body **120**. Stacked body **120** may include first and second planar conductors (e.g., first and second planar conductive layers **121a** and **121b**; hereinafter may be collectively referred to as planar conductive layers **121**) disposed one above the other and substantially parallel to each other, and a dielectric (e.g., a dielectric layer **122**)

interposed therebetween. In one embodiment, second planar conductive layer **121b** and dielectric layer **122** may be disposed on first conductive layer **121a** to respectively cover only a portion(s) of the upper surface of underlying first conductive layer **121a**, such that first conductive layer **121a** may include one or more upper surface portions that are not covered by overlying second planar conductive layer **121b** and dielectric layer **122** (e.g., first and second upper surface portions **12a** and **12b**). By way of a non-limiting example, one or more upper surface portions **12a** and **12b** may be the end portions of first planar conductive layer **121a**, such that they, together with second planar conductive layer **121b** and first dielectric layer **122**, may form one or more stair steps at one or more sides of capacitor **100**.

[0026] In one embodiment, as shown in FIGS. **1A-1C**, second planar conductive layer **121b** may be smaller than its underlying first planar conductive layer **121a** in at least one of its two extending dimensions. For convenience of description, an xyz coordinate system is shown in FIG. **1A**. The x and z axes in the this coordinate system respectively indicate the two orthogonal directions along which planar conductive layers **121** may extend, and the y axis indicate the direction that is orthogonal to the x and z axes. In the above coordinate system, second planar conductive layer **121b** may be smaller than its underlying first planar conductive layer **121a** in at least one of the two dimensions in the directions of x and y axes shown in FIG. **1A**. Further, in the above embodiment, second planar conductive layer **121b** may be equal to or smaller than its underlying dielectric layer **122** in at least one of its two extending dimensions (e.g., the two dimensions in the direction of x and y axes shown in FIG. **1A**). However, it should be appreciated that a stacked body in accordance with the present disclosure is not limited thereto, and depending on particular embodiments, the second planar conductive layer may be equal to or greater in size in at least one of its two extending dimensions than at least some of its underlying first planar dielectric layers and/or dielectric layer, and still may be arranged in a manner that cover only a portion(s) of the upper surface of the first conductive layer.

[0027] In one embodiment, capacitor **100** may further include first and second contacts **130a** and **130b** (which, hereinafter, may be collectively referred to by example as metal lines **130**) (e.g., metal contacts, poly-silicon contacts, etc.) respectively coupled to the upper surfaces of first and second planar conductive layers **121a** and **121b**, and first and second contact electrodes **140a** and **140b** (hereinafter may be collectively referred to as contact electrodes **140**) respectively disposed on and coupled to first and second metal lines **130a** and **130b**. First and second metal lines **130a** and **130b** may be buried inside an interlayer dielectric layer **150**, and first and second contact electrodes **140a** and **140b** may be disposed on interlayer dielectric layer **150** to be respectively coupled to first and second metal lines **130a** and **130b**.

[0028] By way of a non-limiting example, metal lines **130** may respectively extend in the direction substantially perpendicular to the upper surfaces of planar conductive layers **120** (e.g., in the direction indicated by the y axis shown in FIG. **1A**). It should be appreciated that the term “substantially perpendicular,” as used herein, includes, but is not limited to, a range of from -30 to $+30$ degrees around the perpendicular direction. While metal lines **130** are illustrated as an elongated rectangular structure, they may take one of a variety of different shapes. For example, metal lines **130** may be, but are not limited to, a cylinder or a tapered column.

[0029] The aforementioned elements of capacitor **100** may respectively be made of various different materials. For example, substrate **110** may be fabricated from one or more materials, which include, but are not limited to, sapphire, glass, or semiconductor materials (e.g., silicon (Si), germanium (Ge), and gallium arsenide (GaAs)). Planar conductive layers **121** and dielectric layer **122** may be respectively made of a conductive material (e.g., polysilicon) and an oxide of the conductive material (e.g., silicon oxide) in any available manner. Metal lines **130** and contact electrodes **140** may respectively be made of tungsten and aluminum. Interlayer dielectric layer **150** may be made of borophosphosilicate glass (BPSG). It should be appreciated, however, that the aforementioned materials are given for illustrative purposes only, and other materials may be used

as appropriate depending on each implementation.

[0030] In the embodiment described with reference to FIGS. 1A-1C, planar conductive layers **121** and dielectric layer **122** are arranged in a way such that first conductive layer **121a** includes one or more portions (e.g., first and second upper surface portions **12a** and **12b**) not covered by overlying second conductive layer **121b** and dielectric layer **122**, so as to allow forming of one or more contacts (e.g., first metal line **130a**) and one or more contact electrodes (e.g., first contact electrode **140a**) from top-down. The aforementioned arrangement not only allows its easy fabrication using one or more of existing semiconductor fabrication techniques (e.g., concurrently with other semiconductor devices, such as an array of three-dimensional non-volatile memory cells), but also allows attaining high capacitance, to a level hitherto not possible, by, for example, stacking similarly configured additional planar conductive layers and dielectric layers therein.

[0031] In this regard, FIGS. 2A-2D show another illustrative embodiment of a capacitor. FIG. 2A shows a perspective view of another illustrative embodiment of a capacitor. FIG. 2B shows a cross-sectional view of the capacitor shown in FIG. 2A taken along line B-B'. FIG. 2C shows a planar view of the capacitor shown in FIG. 2A. FIG. 2D shows a schematic circuit diagram of the capacitor shown in FIG. 2A. Referring to FIGS. 2A-2D, a capacitor **200** may include a substrate **210**, a stacked body **220**, a plurality of contacts **230a-230d** (which, hereinafter, may be collectively referred to by example as metal lines **230**), first and second contact electrodes **240a** and **240b** (which hereinafter may be collectively referred to as contact electrodes **240**), and an interlayer dielectric layer **250**. While not expressly illustrated in FIGS. 2A-2D for the sake of simplicity, a dielectric material, such as silicon oxide, may be interposed between substrate **210** and stacked body **220**. Stacked body **220** may include N planar conductors (e.g. N planar conductive layers **221a-221d**, which hereinafter may be collectively referred to as planar conductive layers **221**) respectively disposed one above the other and N-1 dielectrics (e.g. N-1 dielectric layers **222a-222c**, which hereinafter may be collectively referred to as dielectric layers **222**) respectively interposed between two adjacent conductive layers **221**. While stacked body **220** of the illustrative embodiment shown in FIGS. 2A-2D include four planar conductive layers **221** and three dielectric layers **222** (i.e., N is equal to 4), a stacked body in accordance with the present disclosure is not limited thereto, and may include any number of planar conductive layers and dielectric layers (e.g., N may be equal to a natural number equal to or greater than 2). It should be noted that numerals in FIGS. 2A-2D that are similar to those in FIGS. 1A-1C generally identify similar components, and unless context dictates otherwise, the descriptions provided with reference to FIGS. 1A-1C generally apply to corresponding components in FIGS. 2A-2D.

[0032] Planar conductive layers **221** and dielectric layers **222** may be alternately disposed one above the other to respectively cover only a portion(s) of the upper surfaces of its underlying planar conductive layers **221**, such that each of planar conductive layers **221** may include at least one portion (e.g. upper surface portions **22a-22f**) not covered by its overlying planar conductive layers **221** and dielectric layers **222**. By way of a non-limiting example, the at least one upper surface portion may be an end portion of one of planar conductive layers **221**, such that the upper surface portions collectively form one or more stair steps at one or more sides of stacked body **220** of capacitor **200** (e.g., the two steps respectively at the right and left sides of stacked body **220** viewed in the direction of the y axis shown in FIG. 2A).

[0033] In one embodiment, as shown in FIGS. 2A-2C, each of planar conductive layers **221** may be smaller than all of its underlying planar conductive layers **221** in at least one of its two extending dimensions (e.g., the two dimensions in the direction of x and y axes shown in FIG. 2A). Further, in the above embodiment, each of planar conductive layers **221** may be equal to or smaller than all of its underlying planar dielectric layers **222** in at least one of its two extending dimensions (e.g., the two dimensions in the direction of x and y axes shown in FIG. 2A). However, it should be appreciated that a stacked body in accordance with the present disclosure is not limited thereto, and depending on particular embodiments, may include one or more planar conductive layers that are

equal to or greater in size than all or some of its underlying planar dielectric layers and/or dielectric layers in at least one of its two extending dimensions.

[0034] One set of the plurality of contacts **230** (e.g., metal lines **230a** and **230c**) may be disposed at one side of stacked body **220** of capacitor **200** (e.g., the left side of stacked body **220** viewed in the direction of the y axis shown in FIG. 2A) and may be respectively coupled to the upper surface portions of alternate ones of planar conductive layers **221** (e.g., odd-numbered planar conductive layers **221**) that are not covered by respective overlying planar conductive layers **221** and dielectric layers **222** (e.g., upper surface portions **22a** and **22e** of planar conductive layers **221a** and **221c**). Further, another set of the plurality of contacts **230** (e.g., metal lines **230b** and **230d**) may be disposed at another side of stacked body **220** of capacitor **200** (e.g., the right side of stacked body **220** viewed in the direction of the y axis shown in FIG. 2A) and may be respectively coupled to one of the upper surface portions of the remaining ones of the planar conductive layers (e.g., even-numbered planar conductive layers **221**), some of which are not covered by respective overlying planar conductive layers **221** and dielectric layers **222** (e.g., upper surface portions **22d** of planar conductive layers **221b**). First contact electrode **240a** may be disposed on and coupled to the one set of the plurality of contacts lines **230**, whereas second contact electrode **240b** may be disposed on and coupled to the another set of the plurality of contacts **230**. Contacts **230** may be buried inside interlayer dielectric layer **250**, and contact electrodes **240** may be disposed on interlayer dielectric layer **250** to be coupled to contacts **230**.

[0035] As can be seen from FIG. 2D, the aforementioned arrangement of stacked body **220** is the equivalent of N-1 capacitors connected in parallel between two nodes A and B respectively corresponding to contact electrodes **240a** and **240b**. In the example shown in FIG. 2D, C1, C2, and C3 respectively represent capacitances provided between planar conductive layers **221a** and **221b**, planar conductive layers **221b** and **221c**, and planar conductive layers **221c** and **221d**. Between each pair of conductive layers, one capacitance is provided at the center portions of the planar conductive layer pair and another capacitance is provided at the end portions of the planar conductive layer pair. In some embodiments, the total capacitance provided by stacked body **220** may be expressed by Equation 1 shown below.

$$C_{\text{total}} = C_{\text{center}} + C_{\text{side}} \quad \text{[Equation 1]}$$

$$= \beta * (W * L * P_n + W * a * (S - 2) * (S - 1))$$

where, Ctotal is the total capacitance provided by stacked body **220**, Ccenter is the capacitance provided between the center portions of planar conductive layers **221** in stacked body **220**, Cside is the capacitance provided between the end portions of planar conductive layers **221** in stacked body **220**, β is the unit capacitance between a planar conductive layer pair, W is the length of planar conductive layers **221** along the y axis shown in FIG. 2A, L is the length of planar conductive layer **221d** along the x axis shown in FIG. 2A, Pn is the number of dielectric layers **222** (i.e., N-1), S is the number of planar conductive layers **221** (i.e., N), and a is the length of an upper surface portion **22a-22d** along the x axis shown in FIG. 2A. As can be appreciated from FIG. 2D and Equation 1 shown above, the total capacitance provided by capacitor **200** is proportional to and thus may be increased by stacking similarly configured additional planar conductive layers and dielectric layers in its stacked body **220**.

[0036] In the illustrative embodiment shown in FIGS. 2A-2D, contact electrodes **240a** and **240b** are disposed above two stair steps located at two opposing ends of capacitor **200**. This is the same for the illustrative embodiment shown in FIGS. 1A-1C. It should be appreciated, however, that contact electrodes in accordance with the present disclosure (and contacts to be coupled thereto) may be disposed in a variety of different ways.

[0037] In this regard, FIGS. 3A-3D show another illustrative embodiment of a capacitor. FIG. 3A shows a perspective view of yet another illustrative embodiment of a capacitor. FIG. 3B shows a cross-sectional view of the capacitor shown in FIG. 3A taken along line C-C'. FIG. 3C shows a

cross-sectional view of the capacitor shown in FIG. 3A taken along line D-D'. FIG. 3D shows a planar view of the capacitor shown in FIG. 3A. Referring to FIGS. 3A-3D, a capacitor **300** may include a substrate **310**, a stacked body **320**, a plurality of contacts **330a-330d** (which, hereinafter, may be collectively referred to by example as metal lines **330**), first and second contact electrodes **340a** and **340d** (which hereinafter may be collectively referred to as contact electrodes **340**), and an interlayer dielectric layer **350**. Stacked body **320** may include N planar conductors (e.g. N planar conductive layers **321a-321d**, which hereinafter may be collectively referred to as planar conductive layers **321**, where N is a natural number greater than or equal to 2) respectively disposed one above the other and N-1 dielectrics (e.g. N-1 dielectric layers **322a-322c**, which hereinafter may be collectively referred to as dielectric layers **322**) respectively interposed between two adjacent conductive layers **321a-321d**. While not expressly illustrated in FIGS. 3A-3D for the sake of simplicity, a dielectric material, such as silicon oxide, may be interposed between substrate **310** and stacked body **320**. Numerals in FIGS. 3A-3D similar to those in FIGS. 1A-1C and 2A-2D generally identify similar components, and unless context dictates otherwise, the descriptions provided with reference to FIGS. 1A-1C and 2A-2D generally apply to corresponding components in FIGS. 3A-3D. For the sake of simplicity, some of the features of capacitor **300** that are similar to those of capacitors **100** and **200** may not be described in the ensuing descriptions.

[0038] In one embodiment, planar conductive layers **321** and dielectric layers **322** may be alternately disposed one above the other to respectively cover end portions at one side of each upper surface of its underlying planar conductive layers **321**, such that each of planar conductive layers **321** may include an end portion at the opposing side of its upper surface (e.g. upper surface portions **32a-32c**) that is not covered by its overlying planar conductive layers **321** and dielectric layers **322**, while the end portion on the one side of its upper surface is completely covered by its overlying planar conductive layers **321** and dielectric layers **322**. In the above embodiment, planar conductive layers **321** and dielectric layers **322** collectively form only one stair step at the one side of stacked body **320** of capacitor **300** (e.g., the step at the left side of capacitor **300** viewed in the direction of the y axis shown in FIG. 3A), as opposed to two stair steps along the both sides of each of stacked bodies **120** and **220** in capacitors **100** and **200** of FIGS. 1A-1C and 2A-2D.

[0039] In the above embodiment, one set of the plurality of contacts **330** (e.g., metal lines **330a** and **330c**) may be disposed adjacent to one lateral side of the step formed by stacked body **320** (e.g., the direction along the x axis shown in FIG. 3A) and coupled to the upper surface portions of a first set of conductors, such as odd-numbered planar conductive layers **321** (e.g., upper surface portions **32a** and **32c** of planar conductive layers **321a** and **321c**). Further, another set of the plurality of contacts **330** (e.g., metal lines **330b** and **330d**) may be disposed adjacent to the other lateral side of the step formed by stacked body **320** (e.g., the direction along the x axis shown in FIG. 3A) and coupled to the upper surface portions of a second set of conductors, such as even-numbered planar conductive layers **321**, some of which are not covered by its overlying planar conductive layers **321** and dielectric layers **322** (e.g., upper surface portions **32d** of planar conductive layers **321b**).

[0040] Further, first and second contact electrodes **340a** and **340b** may both be disposed along a direction substantially parallel to the lateral sides of the step formed on stacked body **320** (e.g., the direction along the x axis shown in FIG. 3A) and be spaced apart from each other by a prescribed distance in a direction perpendicular to the lateral sides of the step formed on stacked body **320** (e.g., the direction along the y axis shown in FIG. 3A), such that first contact electrode **340a** may be coupled to a set of contacts coupled to the upper surface portions of odd-numbered planar conductive layers **321**, while second contact electrode **340b** may be coupled to another set of contacts coupled to the upper surface portions of even-numbered planar conductive layers **321**. Contacts **330** may be buried inside interlayer dielectric layer **350**, and contact electrodes **340** may be disposed on interlayer dielectric layer **350** to be coupled to contacts **330**.

[0041] In the illustrative embodiment shown in FIGS. 1A-1C, 2A-2D and 3A-3D, capacitors **100-300** each include two contact electrodes. It should be appreciated, however, that capacitors in

accordance with the present disclosure may include three or more contact electrodes.

[0042] In this regard, FIGS. 4A and 4B show a planar view of illustrative embodiments of capacitors including four contact electrodes. Referring to FIGS. 4A and 4B, each of capacitors **401** and **402** include four contact electrodes (i.e., contact electrodes **441a-441d** and **442a-442d**) that are disposed in a direction perpendicular to the lateral sides of the steps formed in capacitors **401** and **402** (e.g., the direction along the x axis shown in FIGS. 4A and 4B). Two contact electrodes that are disposed on contacts coupled to a first set of planar conductors, such as odd-numbered planar conductive layers (e.g., contact electrodes **441a** and **441c** coupled to contacts **431a** in FIG. 4A, and contact electrodes **442a** and **442c** coupled to contacts **432a** in FIG. 4B) are alternately arranged with the other two contact electrodes that are disposed on contacts coupled to a second set of planar conductors, such as even-numbered planar conductive layers (e.g., contact electrodes **441b** and **441d** coupled to contacts **431b** in FIG. 4A, and contact electrodes **442b** and **442d** coupled to contacts **432b** in FIG. 4B). In FIG. 4A, contacts **431a** and **431b** are disposed on only one of the two side of capacitor **401** viewed in the direction perpendicular to the lateral sides of the steps thereon (e.g., each of the left and right sides of capacitor **401** viewed in the direction of the y axis shown in FIG. 4A). In FIG. 4B, however, contacts **432a** and **432b** are disposed on both sides of capacitor **402** viewed in the direction perpendicular to the lateral sides of the steps thereon (e.g., both sides of capacitor **402** viewed in the direction of the y axis shown in FIG. 4B). The arrangements of contact electrodes shown in FIGS. 4A and 4B may provide, further to the capacitance provided by their stacked bodies, additional capacitance between the contact electrodes.

[0043] The capacitors described in conjunction with the preceding figures may be fabricated into a variety of semiconductor devices to be used as a passive circuit element therein. Especially, by virtue of their structural configurations hitherto described, capacitors in accordance with the present disclosure may be fabricated concurrently with other semiconductor elements, such as a three-dimensional memory cell array structure of a flash memory device. In this regard, FIG. 5 shows a cross-sectional view of an illustrative embodiment of a flash memory device including a capacitor in accordance with the present disclosure. Referring to FIG. 5, a flash memory device **500** may include a memory cell array region **51** and a periphery region **52**.

[0044] Memory cell array region **51** may include a three-dimensional memory cell array structure **501**. Three-dimensional memory cell array structure **501** may include a substrate **560**, a dielectric layer **561** located on substrate **560**, and a stacked body **570** located on dielectric layer **561** and alternately stacked with planar conductors (e.g. planar conductive layers **571a-571d**) and dielectrics (e.g. dielectric layers **572a-572c**). Stacked body **570** may include one or more pillar-shaped semiconductor structures (e.g., a pillar-shaped semiconductor structure **56**) that may respectively function as a string of three dimensional flash memory cells. Each pillar-shaped memory structure may include, for example, a silicon pillar (e.g., a silicon pillar **57**) and an oxide-nitride-oxide (ONO) film (e.g., an ONO film **58**) encircling the silicon pillar. Each planar conductive layer **571a-571d** functions as a word line for controlling the portion of pillar-shaped memory structure **56** it encircles. For example, the portion of ONO film **57** surrounded by planar conductive layer **571a** may function as a transistor that turns on and off depending on the voltage applied by planar conductive layer **571a** functioning as a word line thereto. Each planar conductive layer **571a-571c** is respectively connected to contact electrodes **590a-590c** through contacts **580a-580c** formed in a dielectric layer **599** to be supplied with program and other types of voltages. The concrete configurations of a three-dimensional memory cell array structure are well known in the pertinent art, and are not further described for the sake of simplicity.

[0045] Periphery region **52** may be formed with a variety of structures/circuits for operating three-dimensional memory cell array structure **501**. For example, periphery region **52** may include one or more capacitors in accordance with the present disclosure to supply necessary voltages to three-dimensional memory cell array structure **501** and/or other parts of flash memory device **500**. In this regard, FIG. 5 shows a portion **502** of such a capacitor. The capacitor may include a substrate **510**,

a dielectric layer **511** located on substrate **510**, and a stacked body **520** located on dielectric layer **511** and alternately stacked with planar conductors (e.g. planar conductive layers **521a-521d**) and dielectrics (e.g. dielectric layers **522a-522c**). Each planar conductive layer **521a-521c** is respectively connected to one of multiple contact electrodes (e.g., a contact electrode **540**) through one of contacts **530** (e.g., contacts **530a** and **530b**) formed in an dielectric layer **550** to be supplied with a charging voltage. As can be appreciated from FIG. 5, the configuration of the three-dimensional memory cell array structure and the capacitor have similar structural configuration, thus, the capacitor may be fabricated in conjunction with and/or concurrently with the three-dimensional memory cell array structure in the memory cell array region. Further, the above similarity allows fabrication of the capacitor, for example, by using the structure(s) that are naturally formed in the peripheral region during the fabrication of the three-dimensional memory cell array structure in the memory cell array region. This will become clearer as we describe an example fabrication process of a capacitor in accordance with this disclosure with regard to FIGS. 7 and 8A-8G.

[0046] The capacitors described in conjunction with the preceding figures may be used for a variety of devices formed in a peripheral region of a semiconductor device. By way of a non-limiting example, capacitors in accordance with the present disclosures may be used as capacitive elements in a charge pump for providing voltages, for example, to contact electrodes **590a-590c** of three-dimensional memory cell array structure **501** shown in FIG. 5. In this regard, FIG. 6 shows an illustrative embodiment of a charge pump including multiple capacitors in accordance with the present disclosure. Referring to FIG. 6, a charge pump **600** may include a plurality of pump stages **610-630** respectively coupled to capacitors **611** and **612**, **621** and **622**, and **631** and **632**. Capacitors **611**, **621**, and **631** may be provided with a clock pulse CLKa, while capacitors **612**, **622**, and **632** may be provided with a clock signal CLKb, which is of the same magnitude as clock signal CLKa but shifted in phase by 180 degrees. The above capacitors may store energy when clock pulse CLKa or CLKb is at Vcc [V], and discharge the energy stored therein when clock pulse CLKa or CLKb is at 0[V]. Each pump stage **610-630** is made of one or more transistors that, when provided with voltage signals discharged from the capacitors, turns on and conveys the voltage signals provided thereto as output. The capacitors in FIG. 6 may provide greater capacitance and store more energy than conventional capacitors (e.g., MOSFET capacitors), while being smaller in size. This allows increasing the voltage output of charge pump **600** without adding an additional pump stage(s) thereto (i.e., without increasing the size and cost of charge pump **600**).

[0047] A method for fabricating a capacitor is explained hereafter with reference to FIGS. 7 and 8A-8G. FIG. 7 shows an example flow diagram of an illustrative embodiment of a method for fabricating a capacitor. Referring to FIG. 7, a substrate may be prepared (block **710**). The substrate, for example, may be prepared by using any of the materials described above with reference to FIGS. 1A-1C (e.g., materials described above with reference to FIG. 1A through FIG. 1C such as, for example, sapphire, glass, or semiconductor materials (e.g., silicon (Si), germanium (Ge), and gallium arsenide (GaAs)). In one embodiment, the substrate may be a substrate for a flash memory device including a memory cell array region and a peripheral region. In block **720**, N planar conductors (e.g. planar conductive layers) are alternately stacked with N-1 dielectrics (e.g. N-1 dielectric layers) on the substrate to form a stacked body thereon. In the embodiment where the substrate is a substrate for a flash memory device, the planar conductive layers and the dielectric layers may be alternately stacked in both the memory cell array region and the peripheral region. In this regard, FIG. 8A shows a cross-sectional view of an illustrative embodiment of portions **819** and **869** of a stacked body respectively formed in a peripheral region **82** and a memory cell array region **81** located on substrates **860** and **810**. In FIG. 8A, the planar conductive layers and the dielectric layers are respectively referenced with numerals **821a-821d** and **822a-822c** (for stacked body portion **820**) and **871a-871d** and **872a-872c** (for stacked body portion **870**). Further, in some embodiments, as shown in FIG. 8A, dielectric layers **861** and **811** may be located on substrates **860**

and **810**, respectively.

[0048] In block **730**, one or more portions of the stacked body are removed to uncover one or more portions of each of the planar conductive layers that were previously covered by their overlying planar conductive and/or dielectric layers. By way of a non-limiting example, the stacked body may be etched with a mask having a width that decreases with each iteration (i.e., a mask slimmed with each iteration) to form a stair step at one or more sides of the stacked body. In the embodiment related to a flash memory device, the portions of the stacked body in both the memory cell array region and the peripheral region may be etched, for example, concurrently, so as to provide two separate stacked bodies in the respective regions. FIG. **8B** shows a cross-sectional view of an illustrative embodiment of portions **820** and **870** of the stacked body respectively iteratively etched with a mask (not shown) having a width that decreases with each iteration (i.e., **W1**, **W2**, and **W3**) to be formed into two separate stacked bodies **820** and **860** each having a stair step. There are various techniques known in the art, including the aforementioned mask slimming technique, for fabricating the aforementioned stair-stepped structure in a stacked body, all of which may be applied to a stacked body of the present disclosure. The technical details thereon are not further described for the sake of simplicity.

[0049] In the embodiment related to a flash memory device, before or after block **730**, the stacked body in the memory cell array region may be processed to form therein one or more pillar-shaped semiconductor structures that may respectively function as a string of three dimensional flash memory cells. Each pillar-shaped memory structure may include, for example, a silicon pillar (e.g., epitaxial silicon or polysilicon) and an oxide-nitride-oxide (ONO) film encircling the silicon pillar. In this regard, FIG. **8C** shows a cross-sectional view of an illustrative embodiment of a pillar-shaped semiconductor structure **86** including a silicon pillar **87** and an ONO film **88**. The techniques for fabricating the aforementioned pillar-shaped semiconductor structure are well known in the pertinent art, and are not further described for the sake of simplicity.

[0050] In block **740**, one or more contacts are formed on the stacked body. The one or more contacts may be substantially perpendicular to the upper surfaces of the stacked body. One set of contacts may be coupled to the uncovered portions of a first set of planar conductors (e.g. odd-numbered planar conductive layers), whereas another set of contacts may be coupled to the uncovered portions of a second set of planar conductors (e.g. even-numbered planar conductive layers).

[0051] In one embodiment, the contacts may be formed by forming an interlayer dielectric layer over the stacked body, and removing one or portions of the interlayer dielectric layer above at least some of the one or more second portions of planar conductive layers to define one or more openings (e.g., holes) therethrough, and depositing conductive materials into the one or more openings to form the contacts therein. In the embodiment related to a flash memory device, one or more contacts may also be formed in the memory cell region. For example, the one or more contacts in the memory cell region may be formed concurrently with the contacts in the periphery region, by also depositing an interlayer dielectric layer in the memory cell region, forming one or more openings in the interlayer dielectric layer, and depositing a conductive material into the openings to form one or more contacts therein. In this regard, FIG. **8D** shows a cross-sectional view of an illustrative embodiment of interlayer dielectric layers **850** and **899** respectively formed in periphery and memory cell array regions **82** and **81**. FIG. **8E** shows a cross-sectional view of an illustrative embodiment of openings **829a** and **829b** formed in periphery region **82** located on substrate **810**, and openings **879a-879c** formed in memory cell array region **81** located on substrate **860**. Further, FIG. **8F** shows a cross-sectional view of an illustrative embodiment of contacts **830a** and **830b** and contacts **880a-880c** respectively formed in periphery and memory cell array regions **82** and **81**.

[0052] In block **750**, two or more contact electrodes are formed on the contacts. For example, a first contact electrode may be formed on a first set of contacts coupled to the odd-numbered planar

conductive layers and a second contact electrode may be coupled to a second set of contacts coupled to even-numbered planar conductive layers. In one embodiment, the first and second electrodes may be formed above the first and second sides of the stacked body at which stair steps are respectively formed. In another embodiment, the first and second contact electrodes may be formed above a stair step adjacent to first and second lateral sides of the stair step, respectively. Further, in yet another embodiment, in addition to the first and second electrodes, additional contact electrodes may be formed. For example, third and fourth contact electrodes may be respectively formed to be disposed on and coupled to at least some of the first set of contacts and at least some of the second set of contacts. The third contact electrode may be interposed between and adjacent to the second and fourth contact electrodes to provide capacitance between at least the second and third contact electrodes or the third and fourth contact electrodes.

[0053] In the embodiment related to a flash memory device, one or more contact electrodes may also be formed in the memory cell region. In this regard, FIG. **8G** shows a cross-sectional view of an illustrative embodiment of a contact electrode **840** formed in peripheral region **82** for a capacitor in accordance with present disclosure and contact electrodes **890a-890c** for the three-dimensional memory cell array formed in memory cell array region **81**.

[0054] As can be appreciated from FIGS. **8A-8G**, a capacitor may be formed by using a portion of a stacked body of alternating planar conductive layers and dielectric layers that is naturally formed in the peripheral region as well as the memory cell array region during the process of fabricating a flash memory device. Further, by virtue of its structural configuration, a capacitor in accordance with the present disclosure may be formed in conjunction with and/or concurrently with the memory cell array structure (e.g., a three-dimensional memory cell array structure) in the memory cell array region.

[0055] FIG. **9** shows a schematic diagram of an illustrative embodiment of a system including a non-volatile memory device (e.g., a flash memory device **500** of FIG. **5**). A system **900** may be used in devices such as, for example, a personal digital assistant (PDA), a laptop or portable computer with wireless capability, a web tablet, a wireless telephone, a pager, an instant messaging device, a digital music player, a digital camera, or other devices that may be adapted to transmit and/or receive information either wirelessly or over a wire connection. The system **900** may be used in any of the following systems: a wireless local area network (WLAN) system, a wireless personal area network (WPAN) system, or a cellular network.

[0056] The system **900** may include a controller **910**, an input/output (I/O) device **920** (e.g., a keypad, display), the flash memory device **500** of FIG. **5**, a wireless interface **940**, and a static random access memory (SRAM) **960** and coupled to each other via a bus **950**. A battery **980** may supply power to the system **900** in one embodiment. The memory device may include a NAND memory, a flash memory, a NOR memory, or the like.

[0057] The controller **910** may include, for example, one or more microprocessors, digital signal processors, micro-controllers, or the like. The flash memory device **500** may be used to store messages transmitted to or by the system **900**. The flash memory device **500** may also optionally be used to store instructions that are executed by controller **920** during the operation of the system **900**, and may be used to store user data either generated, collected or received by the system **900** (such as image data). The instructions may be stored as digital information and the user data, as disclosed herein, may be stored in one section of the memory as digital data and in another section as analog memory. As another example, a given section at one time may be labeled as such and store digital information, and then later may be relabeled and reconfigured to store analog information.

[0058] The I/O device **920** may be used to generate a message. The system **900** may use the wireless interface **940** to transmit and receive messages to and from a wireless communication network with a radio frequency (RF) signal. Examples of the wireless interface **940** may include an antenna, or a wireless transceiver, such as a dipole antenna, although the scope of the present

disclosure is not limited in this respect. Also, the I/O device **920** may deliver a voltage reflecting what is stored as either a digital output (if digital information was stored), or as analog information (if analog information was stored). While an example in a wireless application is provided above, embodiments of the present invention may also be used in non-wireless applications as well.

[0059] It should be appreciated that the structural and functional configurations of a capacitor, a semiconductor device, and/or a system and their elements described in conjunction with FIGS. **1A-9** are indicative of a few ways in which a capacitor, a semiconductor device, and/or a system may be implemented. It should be appreciated that a capacitor in accordance with this disclosure may be applied to any type of devices and systems, including types of memories other than flash memory.

[0060] One skilled in the art will appreciate that, for this and other processes and methods disclosed herein, the functions performed in the processes and methods may be implemented in differing order. Furthermore, the outlined steps and operations are only provided as examples, and some of the steps and operations may be optional, combined into fewer steps and operations, or expanded into additional steps and operations without detracting from the essence of the disclosed embodiments.

[0061] The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. It is to be understood that this disclosure is not limited to particular methods, reagents, compounds compositions or biological systems, which can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

[0062] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

[0063] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one

of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

[0064] In addition, where features or aspects of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

[0065] As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, etc. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third, and upper third, etc. As will also be understood by one skilled in the art all language such as “up to,” “at least,” and the like include the number recited and refer to ranges which can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 cells refers to groups having 1, 2, or 3 cells. Similarly, a group having 1-5 cells refers to groups having 1, 2, 3, 4, or 5 cells, and so forth.

[0066] From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A memory device, comprising: a stack structure including tiers vertically stacked relative to one another and respectively comprising conductive material vertically neighboring dielectric material, the stack structure having at least one stair step structure individually including steps defined by horizontal ends of at least some of the tiers; pillars respectively comprising semiconductor material vertically extending completely through the stack structure; and conductive contacts within a horizontal area of the at least one stair step structure of the stack structure, the conductive contacts individually in physical contact with the conductive material of a respective tier of the stack structure at a respective step of the at least one stair step structure.
2. The memory device of claim 1, wherein: the at least one stair step structure comprises a first stair step structure including a first set of the steps; and a first group of the conductive contacts is within a horizontal area of the first set of the steps, each conductive contact of the first group of the conductive contacts in physical contact with a different step of the first set of the steps than each other conductive contact of the first group of the conductive contacts.
3. The memory device of claim 2, wherein at least one step of the first set of the steps is free of any of the conductive contacts in physical contact therewith.

4. The memory device of claim 3, wherein the at least one step of the first set of the steps is horizontally interposed between at least two other steps of the first set of the steps.
5. The memory device of claim 2, wherein at least two conductive contacts of the first group of the conductive contacts are horizontally offset from one another in a first direction and are substantially horizontally aligned with one another in a second direction orthogonal to the first direction.
6. The memory device of claim 2, wherein at least two conductive contacts of the first group of the conductive contacts are horizontally offset from one another in each of a first direction and a second direction orthogonal to the first direction.
7. The memory device of claim 2, wherein: the at least one stair step structure further comprises a second stair step structure within a vertical span of the first stair step structure and including a second set of the steps; and a second group of the conductive contacts is within a horizontal area of the second set of the steps, each conductive contact of the second group of the conductive contacts in physical contact with a different step of the second set of the steps than each other conductive contact of the second group of the conductive contacts.
8. The memory device of claim 7, wherein each conductive contact of the second group of the conductive contacts terminates at a different vertical position within the stack structure than each conductive contact of the first group of the conductive contacts.
9. The memory device of claim 7, wherein at least two conductive contacts of the second group of the conductive contacts horizontally offset from at least two conductive contacts of the first group of the conductive contacts in a first direction are substantially horizontally aligned with one another and with the at least two conductive contacts of the first group of the conductive contacts in a second direction orthogonal to the first direction.
10. The memory device of claim 1, wherein the pillars individually further comprise each of oxide material, nitride material, and additional oxide material respectively outwardly encircling the semiconductor material thereof.
11. The memory device of claim 1, further conductive routing structures vertically above and within a horizontal area of at least one stair step structure of the stack structure, the conductive routing structures respectively coupled to at least one of the conductive contacts.
12. A non-volatile memory device, comprising: a stack structure including conductive structures vertically alternating with dielectric structures, the stack structure comprising: a memory array region including vertically extending strings of non-volatile memory cells within a horizontal area thereof; and a stair step region horizontally offset from the memory array region in a first direction and including a stair step structure having steps comprising edges of a group of the conductive structures within a vertical span of the vertically extending strings of non-volatile memory cells; and conductive contact structures within a horizontal area of the stair step region of the stack structure, the conductive contact structures individually physically contacting a respective one of the steps of the stair step structure and electrically coupled to circuitry configured to operate the vertically extending strings of non-volatile memory cells.
13. The non-volatile memory device of claim 12, wherein the steps of the stair step structure comprise: a first step having a first of the conductive contact structures thereon; a second step vertically above and directly horizontally adjacent to the first step in the first direction, the second step free of any of the conductive contact structures thereon; and a third step vertically above and directly horizontally adjacent to the second step in the first direction, the third step having a second of the conductive contact structures thereon.
14. The non-volatile memory device of claim 13, wherein the first of the conductive contact structures is substantially horizontally aligned with the second of the conductive contact structures in a second direction orthogonal to the first direction.
15. The non-volatile memory device of claim 12, wherein the steps of the stair step structure comprise: a first step having a first of the conductive contact structures thereon; a second step

vertically above and directly horizontally adjacent to the first step in the first direction, the second step having a second of the conductive contact structures thereon, the second of the conductive contact structures horizontally offset from the first of the conductive contact structures in a second direction orthogonal to the first direction; a third step vertically above and directly horizontally adjacent to the second step in the first direction, the third step having a third of the conductive contact structures thereon, the third of the conductive contact structures horizontally offset from the second of the conductive contact structures in the second direction; and a fourth step vertically above and directly horizontally adjacent to the third step in the first direction, the fourth step having a fourth of the conductive contact structures thereon, the fourth of the conductive contact structures horizontally offset from the third of the conductive contact structures in the second direction.

16. The non-volatile memory device of claim 15, wherein: the third of the conductive contact structures horizontally overlaps the first of the conductive contact structures in the second direction; and the fourth of the conductive contact structures horizontally overlaps the second of the conductive contact structures in the second direction.

17. The non-volatile memory device of claim 12, wherein the stack structure further comprises an additional stair step region horizontally offset from the stair step region and the memory array region in the first direction, the additional stair step region including an additional stair step structure having opposite slope than the stair step structure and comprising additional steps comprising additional edges of the group of the conductive structures within the vertical span of the vertically extending strings of non-volatile memory cells.

18. The non-volatile memory device of claim 17, further comprising additional conductive contact structures within a horizontal area of the additional stair step region of the stack structure, the additional conductive contact structures individually physically contacting a respective one the additional steps of the additional stair step structure.

19. A 3D NAND Flash memory device, comprising: a stack structure including levels of conductive material vertically interleaved with levels of dielectric structures, the stack structure including: a first stair step structure having positive slope and including first steps comprising first horizontal ends of at least some of the levels of conductive material; and a second stair step structure having negative slope and including second steps comprising second horizontal ends of the at least some of the levels of conductive material; first conductive contacts within a horizontal area of the first stair step structure and contacting at least some of the first steps; second conductive contacts within a horizontal area of the second stair step structure and contacting at least some of the second steps; and vertically extending strings of memory cells within a vertical span of the stack structure.

20. The 3D NAND Flash memory device of claim 19, wherein: the second conductive contacts are horizontally offset from the first conductive contacts in a first direction; and at least some of the second conductive contacts horizontally overlap at least some of the first conductive contacts in a second direction perpendicular to the first direction.
